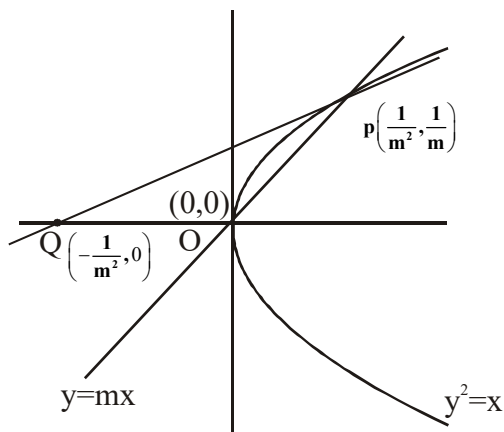


61. Let a line $y = mx$ ($m > 0$) intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area (ΔOPQ) = 4 sq. units, then m is equal to _____.

माना एक रेखा $y = mx$ ($m > 0$), परवलय $y^2 = x$ को मूल बिन्दु के अतिरिक्त एक बिन्दु P पर काटती है। माना P पर इसकी स्पर्श रेखा x- अक्ष को बिन्दु Q पर मिलती है। यदि ΔOPQ का क्षेत्रफल 4 वर्ग इकाई है, तो m बराबर है _____।

Ans. 0.5

Sol.



Equation of tangent at P

$$T = 0$$

$$y \cdot \frac{1}{m} = \left(x + \frac{1}{m^2} \right)$$

$$\frac{2y}{m} = x + \frac{1}{m^2}$$

$$Q\left(-\frac{1}{m^2}, 0\right)$$

$$\text{Area } \Delta OPQ = 4$$

$$\frac{1}{2} \times \frac{1}{m^2} \times \frac{1}{m} = 4$$

$$m = \frac{1}{2} = 0.5$$

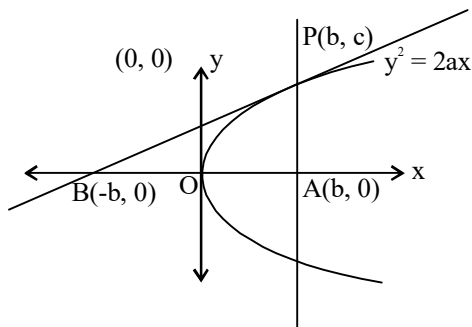
62. Let S be the set of all $a \in \mathbb{N}$ such that the area of the triangle formed by the tangent at the point $P(b,c)$, $b, c \in \mathbb{N}$,

on the parabola $y^2 = 2ax$ and the lines $x = b, y = 0$ is 16 unit^2 , then $\sum_{a \in S} a$ is equal to

माना सभी $a \in \mathbb{N}$ जिनके लिए परवलय $y^2 = 2ax$ के बिंदु $P(b, c), b, c \in \mathbb{N}$ पर स्पर्श रेखा तथा रेखाओं $x = b, y = 0$ से बने त्रिभुज का क्षेत्रफल 16 वर्ग इकाई है, का समुच्चय S है, तो $\sum_{a \in S} a$ बराबर है।

Ans. 146

Sol.



As $P(b, c)$ lies on parabola so $c^2 = 2ab$ _____ (1)

Now equation of tangent to parabola $y^2 = 2ax$ in point form is $yy_1 = 2a \frac{(x + x_1)}{2}, (x_1, y_1) = (b, c)$

$$\Rightarrow yc = a(x + b)$$

For point B, put $y = 0$, now $x = -b$

So, area of $\Delta PBA, \frac{1}{2} \times AB \times AP = 16$

$$\Rightarrow \frac{1}{2} \times 2b \times c = 16$$

$$\Rightarrow bc = 16$$

As b and c are natural number so possible values of (b, c) are $(1, 16), (2, 8), (4, 4), (8, 2)$ and $(16, 1)$

Now from equation (1) $a = \frac{c^2}{2b}$ and $a \in \mathbb{N}$, so value of (b, c) are $(1, 16), (2, 8)$ and $(4, 4)$ now values of a are $128, 16$ and 2 .

Hence sum of values of a is 146

63. The equations of the sides AB, BC and CA of a triangle ABC are : $2x + y = 0, x + py = 21a, (a \neq 0)$ and $x - y = 3$ respectively. Let $P(2, a)$ be the centroid of ΔABC . Then $(BC)^2$ is equal to _____.

एक त्रिभुज ABC की भुजाओं AB, BC तथा CA के समीकरण क्रमशः $2x + y = 0, x + py = 21a, (a \neq 0)$ तथा $x - y = 3$ है। माना ΔABC का केन्द्रक $P(2, a)$ है। तब $(BC)^2$ बराबर है।

Ans. 122

Sol. Now $\frac{1+\alpha+\beta}{3} = 2$

$$\alpha + \beta = 5 \dots (1)$$

$$\text{and } \frac{-2-2\beta+\alpha-3}{3} = a \Rightarrow \alpha - 2\beta = 3a + 5 \quad \dots (2)$$

from (1) and (2) $\alpha = a + 5, \beta = -a$

$$B(-a, 2a), C(a+5, a+2)$$

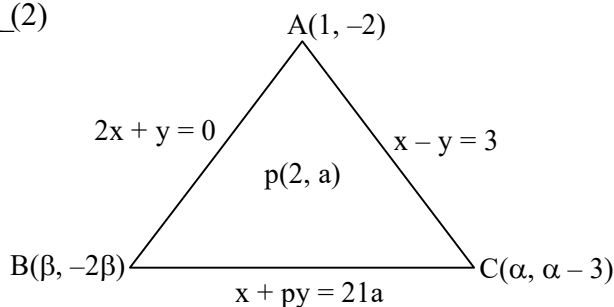
B and C lies on $x + py = 21a$

$$-a + p(2a) = 21a \Rightarrow p = 11$$

$$\text{and } a + 5 + 11(a + 2) = 21a$$

$$\Rightarrow a = 3$$

$$B(-3, 6), C(8, 5), BC^2 = 122$$



64. Let $\tan \alpha, \tan \beta$ and $\tan \gamma; \alpha, \beta, \gamma \neq \frac{(2n-1)\pi}{2}, n \in \mathbb{N}$ be the slopes three line segments OA, OB and OC,

respectively, where O is origin. If circumcentre of $\triangle ABC$ coincides with origin and its orthocentre lies on y-axis, then the value of $\left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$ is equal to _____.

माना तीन रेखाखंडों OA, OB तथा OC (O मूलबिन्दु है) की प्रवणताएँ क्रमशः $\tan \alpha, \tan \beta$ तथा $\tan \gamma; \alpha, \beta, \gamma \neq \frac{(2n-1)\pi}{2},$

$n \in \mathbb{N}$ है। यदि $\triangle ABC$ का परिकेन्द्र मूलबिन्दु है तथा इसका लम्बकेन्द्र y-अक्ष पर है, तो $\left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$ का मान

बराबर _____।

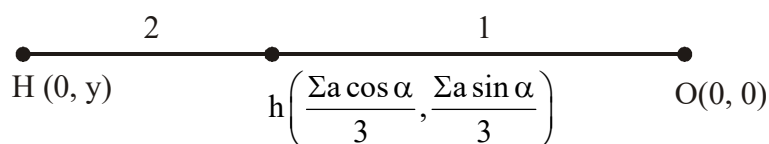
Ans. 144

Sol. Let,

$$A(a \cos \alpha, a \sin \alpha)$$

$$B(a \cos \beta, a \sin \beta)$$

$$C(a \cos \gamma, a \sin \gamma)$$



$$\frac{2 \times 0 + 1 \times 0}{2+1} = \frac{\sum a \cos \alpha}{3}$$

$$\Rightarrow \sum \cos \alpha = 0$$

$$\Rightarrow \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\text{then, } \cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma = 3 \cos \alpha \cos \beta \cos \gamma$$

$$\text{Now, } \left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$$

$$\left(\frac{4(\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma) - 3(\cos \alpha + \cos \beta + \cos \gamma)}{\cos \alpha \cos \beta \cos \gamma} \right)^2$$

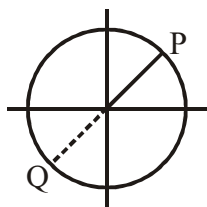
$$= \left(\frac{12 \cos \alpha \cos \beta \cos \gamma}{\cos \alpha \cos \beta \cos \gamma} \right)^2$$

$$= 144$$

65. Let PQ be a diameter of the circle $x^2 + y^2 = 9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line, $x + y = 2$ respectively, then the maximum value of $\alpha\beta$ is _____ .

माना PQ वृत्त $x^2 + y^2 = 9$ का एक व्यास है। यदि P तथा Q से रेखा $x + y = 2$ पर खींचे गए लंबों की लंबाइयाँ क्रमशः α तथा β हैं, तो $\alpha\beta$ का अधिकतम मान है _____ ।

Ans. 7



Sol.

$$P(3\cos\theta, 3\sin\theta)$$

$$Q(-3\cos\theta, -3\sin\theta)$$

$$\alpha = \left| \frac{3\cos\theta + 3\sin\theta - 2}{\sqrt{2}} \right|$$

$$\Rightarrow \alpha\beta_{\max} = \left| \frac{9 \times 2 - 4}{2} \right| = 7$$

$$\alpha\beta = \left| \frac{9(\cos\theta + \sin\theta)^2 - 4}{2} \right|$$

$$(\cos\theta + \sin\theta)_{\max}^2 = 2$$

$$\Rightarrow \alpha\beta_{\max} = \left| \frac{9 \times 2 - 4}{2} \right| = 7$$

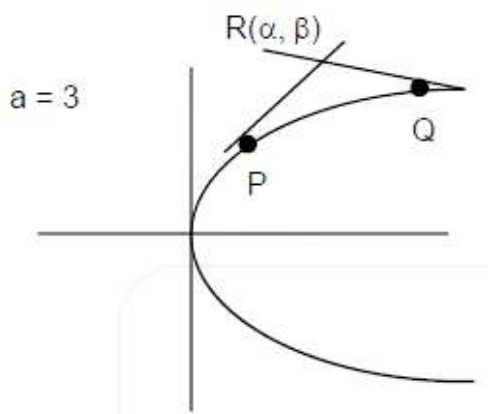
66. The ordinates of the points P and Q on the parabola with focus (3, 0) and directrix $x = -3$ are in the ratio 3 : 1. If $R(\alpha, \beta)$ is the point of intersection of the tangents to the parabola at P and Q, then $\frac{\beta^2}{\alpha}$ is equal to _____.

परवलय जिसकी नाभि (3, 0) तथा नियता $x = -3$ हैं, कि बिंदुओं P तथा Q की कोटियाँ 3 : 1 के अनुपात में हैं। यदि P तथा Q पर परवलय की स्पर्श रेखाओं का प्रतिच्छेदन बिंदु $R(\alpha, \beta)$ है, तो $\frac{\beta^2}{\alpha}$ बराबर है :

Ans. 16

Sol. Parabola is

$$y^2 = 12x$$



$$P(at_1^2 \cdot 2at_1), \quad Q(at_2^2 \cdot 2at_2) \quad \alpha = at_1t_2 \quad \text{and} \quad \beta = a(t_1 + t_2)$$

$$2at_2 = 3 \cdot 2at_1$$

$$t_2 = 3t_1$$

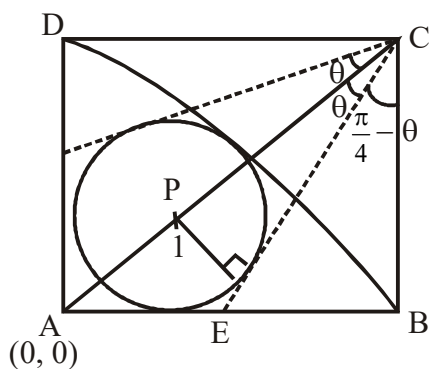
$$\frac{\beta^2}{\alpha} = \frac{a^2(t_1 + t_2)^2}{att_1t_2} = \frac{a(16t_1^2)}{3t_1^2} = 16$$

67. Let ABCD be a square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines AD and AB are tangent to it, is also drawn. Let a tangent line from the point C to the circle C_2 meet the side AB at E. If the length of EB is $\alpha + \sqrt{3} \beta$, where α, β are integers, then $\alpha + \beta$ is equal to _____.

माना भुजा की इकाई लम्बाई एक वर्ग ABCD है। माना इकाई त्रिज्या तथा केन्द्र A का एक वृत्त C_1 खींचा जाता है। वृत्त C_1 तथा रेखाओं AD और AB को स्पर्श करता हुआ एक और वृत्त C_2 भी खींचा जाता है। माना बिन्दु C से वृत्त C_2 की एक स्पर्श रेखा भुजा AB को E पर मिलती है। यदि EB की लम्बाई $\alpha + \sqrt{3} \beta$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है _____।

Ans. 1

Sol.



Let centre of $C_2 = P(r, r)$

$$\sqrt{2} r = |1 - r|$$

$$\sqrt{2} r = 1 - r$$

$$(\sqrt{2} + 1)r = 1$$

$$r = \sqrt{2} - 1$$

$$PC = AC - AP$$

$$PC = AC - \sqrt{2} r$$

$$= \sqrt{2} - \sqrt{2} (\sqrt{2} - 1)$$

$$PC = 2(\sqrt{2} - 1)$$

$$\sin \theta = \frac{r}{PC} = \frac{\sqrt{2} - 1}{2(\sqrt{2} - 1)} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\frac{BE}{BC} = \tan\left(\frac{\pi}{4} - \theta\right)$$

$$BE = \tan(15^\circ) = 2 - \sqrt{3}$$

$$\alpha = 2, \beta = -1$$

$$\alpha + \beta = 1$$

68. Let $H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in \mathbb{N}$, Let k be the smallest even value of n such that the eccentricity of H_k is a rational number. If l is the length of the latus rectum of H_k , then $21l$ is equal to _____.

माना $H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in \mathbb{N}$ है। माना k, n का वह न्यूनतम सम मान है जिसके लिए H_k की उत्केन्द्रता एक परिमेय संख्या है। यदि H_k की नाभिलंब जीवा की लंबाई l है, तो $21l$ बराबर है।

Ans. 306

Sol. $(3+n) = (1+n)(e^2-1)$

$$e^2 = \frac{2n+4}{n+1} = \frac{2(n+2)}{n+1}$$

Check when $(n+1) = 9, 25, 49, \dots$

$$n=8, e^2 = \frac{20}{9}$$

$$n=24, e^2 = \frac{52}{25}$$

$$n=48, e^2 = \frac{100}{49} \Rightarrow e = \frac{10}{7}$$

$$\Rightarrow n=48$$

$$\Rightarrow 21l = 21 \times \frac{2b^2}{a} = 42 \times \frac{n+3}{\sqrt{n+1}} = \frac{42 \times 51}{7} = 306$$

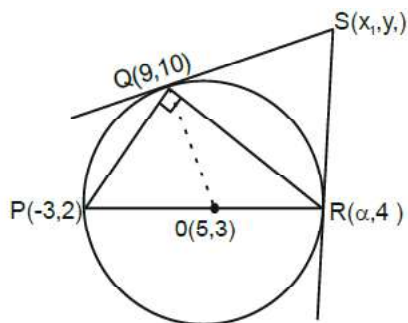
69. Points $P(-3,2)$, $Q(9,10)$ and $R(\alpha, 4)$ lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point. If S lies on the line $2x - ky = 1$, then k is equal to _____.

बिंदु $P(-3,2)$, $Q(9,10)$ तथा $R(\alpha, 4)$ एक वृत्त पर है, जिसका एक व्यास PR है। बिंदुओं Q तथा R पर वृत्त C की स्पर्श रेखाएँ बिंदु S पर मिलती हैं। यदि बिंदु S रेखा $2x - ky = 1$ पर है, तो k बराबर है।

Ans. 3

$$\Rightarrow \frac{10-2}{9+3} \times \frac{10-4}{9-\alpha} = -1 \Rightarrow \alpha = 13$$

$$m_{0P} \cdot m_{QS} = -1 \Rightarrow m_{QS} = -\frac{4}{7}$$


$$y - 10 = -\frac{4}{7}(x - 9)$$

$$\Rightarrow 4x + 7y = 106 \quad \dots(1)$$

$$m_{\text{OR}} - m_{\text{RS}} = -1 \Rightarrow m_{\text{RS}} = -8$$

$$y - 4 = -8(x - 13)$$

$$\Rightarrow 8x + y = 108 \quad \dots\dots(2)$$

Solving eq. (1) & (2)

$$x_1 = \frac{25}{2} y_1 = 8$$

$S(x_1, y_1)$ lies on $2x - ky = 1$

$$25 - 8k = 1$$

$$\Rightarrow 8k = 24$$

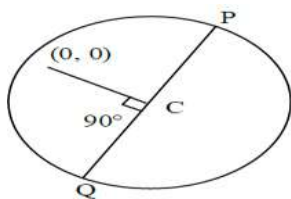
$$\Rightarrow k = 3$$

70. Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C(\sqrt{2}, \sqrt{3})$. Let O be the origin and OC be perpendicular to both CP and CQ . If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + b_1^2 + a_2^2 + b_2^2$ is equal to _____.

माना वृत्त $C(\sqrt{2}, \sqrt{3})$ केन्द्र के एक वृत्त पर दो भिन्न बिंदु $P(a_1, b_1)$ एवं $Q(a_2, b_2)$ है। माना O मूल बिंदु है तथा OC , रेखाओं CP एवं CQ दोनों पर लम्बवत् है। यदि त्रिभुज OCP का क्षेत्रफल $\frac{\sqrt{35}}{2}$ है, तो $a_1^2 + b_1^2 + a_2^2 + b_2^2$ बराबर है।

Ans. 24

Sol. $\frac{1}{2} \times PC \times \sqrt{5} = \frac{\sqrt{35}}{2}; PC = \sqrt{7}$



$$a_1^2 + b_1^2 + a_2^2 + b_2^2 = OP^2 + OQ^2$$

$$= 2(5 + 7) = 24$$

71. If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to _____.

यदि रेखा $l_1 : 3y - 2x = 3$ रेखाओं $l_2 : x - y + 1 = 0$ तथा $l_3 : \alpha x + \beta y + 17 = 0$ की कोण समद्विभाजक है, तो $\alpha^2 + \beta^2 - \alpha - \beta$ बराबर है _____

Ans. 348

Sol. Point of intersection of $l_1 : 3y - 2x = 3$

$l_2 : x - y + 1 = 0$ is $P \equiv (0, 1)$

Which lies on $l_3 : \alpha x + \beta y + 17 = 0$

$$\Rightarrow \beta = -17$$

Consider a random point $Q \equiv (-1, 0)$

on $l_2 : x - y + 1 = 0$, image of Q about

$l_2 : x - y + 1 = 0$, is $Q' \equiv \left(\frac{-17}{13}, \frac{6}{13} \right)$ which is calculated by formulae

$$\frac{x - (-1)}{2} = \frac{y - 0}{-3} = 2 \left(\frac{-2 + 3}{13} \right)$$

Now, Q' lies in $l_3 : \alpha x + \beta y + 17 = 0$

$$\Rightarrow \alpha = 7$$

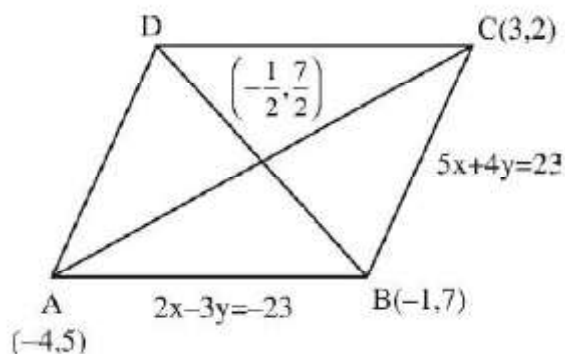
$$\text{Now, } \alpha^2 + \beta^2 - \alpha - \beta = 348$$

72. Let the equations of two adjacent sides of a parallelogram ABCD be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d, then $50d^2$ is equal to _____.

माना एक समांतर चतुर्भुज की दो संलग्न भुजाओं के समीकरण $2x - 3y = -23$ तथा $5x + 4y = 23$ हैं। यदि इसके एक विकर्ण AC का समीकरण $3x + 7y = 23$ है तथा A की दूसरे विकर्ण से दूरी d है, तो $50d^2$ बराबर है :

Ans. 529

Sol.



A & C point will be $(-4, 5)$ & $(3, 2)$

mid point of AC will be $\left(-\frac{1}{2}, \frac{7}{2}\right)$

equation of diagonal BD is

$$y - \frac{7}{2} = \frac{\frac{7}{2} - 5}{-\frac{1}{2} - 3} \left(x + \frac{1}{2}\right)$$

$$\Rightarrow 7x + y = 0$$

Distance of A from diagonal BD

$$= d = \frac{23}{\sqrt{50}}$$

$$\Rightarrow 50d^2 = (23)^2$$

$$50d^2 = 529$$

73. A triangle is formed by the tangents at the point $(2, 2)$ on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to _____.

वक्रों $y^2 = 2x$ तथा $x^2 + y^2 = 4x$ के बिंदु $(2, 2)$ पर स्पर्श रेखाएँ तथा रेखा $x + y + 2 = 0$ एक त्रिभुज बनाती है। यदि इस त्रिभुज के परिवृत्त की त्रिज्या r है, तो r^2 बराबर है।

Ans. 10

Sol. $S_1 : y^2 = 2x$

$S_2 : x^2 + y^2 = 4x$

$P(2,2)$ is common point on S_1 & S_2

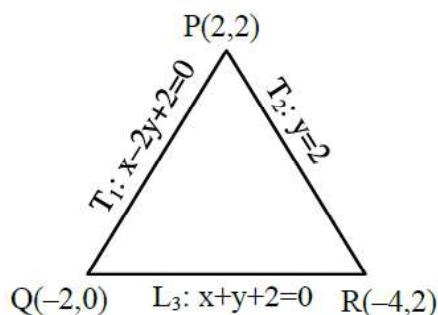
T_1 is tangent to S_1 at $P \Rightarrow T_1 : y \cdot 2 = x + 2$

$$\Rightarrow T_1 : x - 2y + 2 = 0$$

T_2 is tangent to S_2 at $P \Rightarrow T_2 : x \cdot 2 + y \cdot 2 = 2(x + 2)$

$$\Rightarrow T_2 : y = 2$$

& $L_3 : x + y + 2 = 0$ is third line



$$PQ = a = \sqrt{20}$$

$$QR = b = \sqrt{8}$$

$$RP = c = 6$$

$$\text{Area } (\Delta PQR) = \Delta = \frac{1}{2} \times 6 \times 2 = 6$$

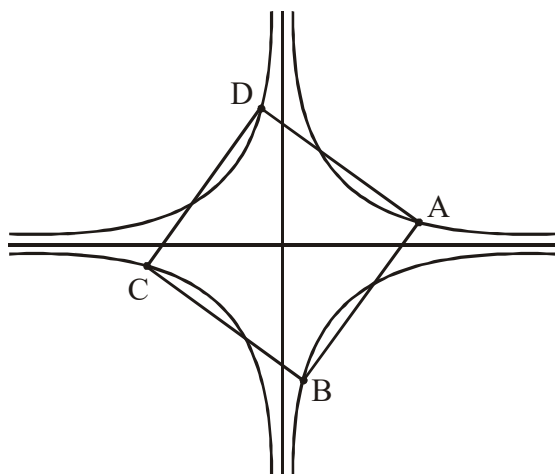
$$\therefore r = \frac{abc}{4\Delta} = \frac{\sqrt{160}}{4} = \sqrt{10} \Rightarrow r^2 = 10$$

74. A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is :

एक वर्ग ABCD के सभी शीर्ष $x^2y^2 = 1$ पर हैं। इसकी भुजाओं के मध्यबिन्दु भी इसी वक्र पर हैं तो ABCD के क्षेत्रफल वर्ग है _____।

Ans. 80

Sol. The curves are $xy = 1$ or $xy = -1$



As per the diagram

$$\text{Let } A \left(t, \frac{1}{t} \right) \quad C \left(-t, \frac{-1}{t} \right)$$

$$B \left(\frac{1}{t}, -t \right) \quad D \left(-\frac{1}{t}, t \right)$$

Mid-point of AB lies on $xy = 1$

$$\Rightarrow \left(\frac{t + \frac{1}{t}}{2} \right) \times \left(\frac{\frac{1}{t} - t}{2} \right) = 1$$

$$\Rightarrow \frac{1}{t^2} - t^2 = 4 \Rightarrow \left(t^2 + \frac{1}{t^2} \right)^2 = \left(t^2 - \frac{1}{t^2} \right)^2 + 4 = 20$$

$$\Rightarrow t^4 + 4t^2 - 1 = 0.$$

$$\begin{aligned} \text{Side of square } AB &= \sqrt{\left(t - \frac{1}{t} \right)^2 + \left(\frac{1}{t} + t \right)^2} \\ &= \sqrt{2 \left(t^2 + \frac{1}{t^2} \right)} \end{aligned}$$

$$\text{Area of square } 2 \times \left(t^2 + \frac{1}{t^2} \right) = 2 \times \sqrt{20}$$

$$\text{Square of Area } 4 \times 20 = 80$$

75. Let an ellipse with centre $(1, 0)$ and latus rectum of length $\frac{1}{2}$ have its major axis along x-axis. If its minor

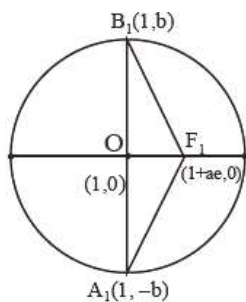
axis subtends an angle 60° at the foci, then the square of the sum of the lengths of its minor and major axes is equal to _____.

यदि दीर्घवृत्त का केन्द्र $(1, 0)$ तथा नाभि लम्ब की लम्बाई $\frac{1}{2}$ है कि दीर्घ x-अक्ष है। यदि लघु अक्ष द्वारा नाभि पर बनाया गया कोण 60°

तो इसकी लघु अक्ष तथा दीर्घ अक्ष की लम्बाइयों के योग का वर्ग होगा _____

Ans. 9

Sol.



$$\text{L.R.} = \frac{2b^2}{a} = \frac{1}{2}$$

$$4b^2 = a \quad \dots(i)$$

$$\text{Ellipse } \frac{(x-1)^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$m_{B_1F_1} = \frac{1}{\sqrt{3}}$$

$$\frac{b}{ae} = \frac{1}{\sqrt{3}}$$

$$3b^2 = a^2e^2 = a^2 - b^2$$

$$4b^2 = a^2 \quad \dots(ii)$$

from (i) and (ii)

$$a = a^2$$

$$\therefore a = 1$$

$$b^2 = \frac{1}{4}$$

$$((2a) + (2b))^2 = 9$$

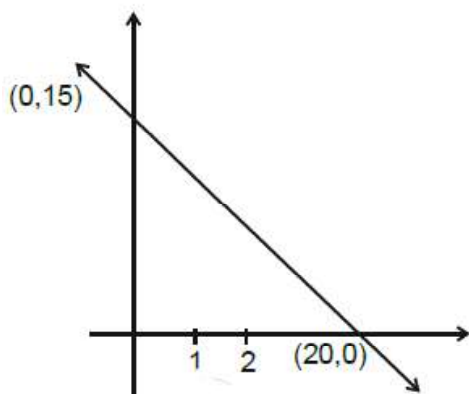
76. A triangle is formed by X-axis, Y-axis and the line $3x + 4y = 60$. Then the number of points $P(a,b)$ which lie

strictly inside the triangle, where a is an integer and b is a multiple of a , is _____.

X-अक्ष, Y-अक्ष तथा रेखा $3x + 4y = 60$ एक त्रिभुज बनाते हैं। तो ऐसे बिंदुओं $P(a,b)$ जहाँ a एक पूर्णांक है तथा b , a का एक गुणज है, जो त्रिभुज के अंदर है, की संख्या है।

Ans. 31

Sol. If $x = 1, y = \frac{57}{4} = 14.25$



$(1, 1) (1, 2) - (1, 14) \Rightarrow 14 \text{ pts.}$

If $x = 2, y = \frac{27}{2} = 13.5$

$(2, 2) (2, 4) ; K. (2, 12) \Rightarrow 6 \text{ pts.}$

If $x = 3, y = \frac{51}{4} = 12.75$

$(3, 3) (3, 6) - (3, 12) \Rightarrow 4 \text{ pts.}$

If $x = 4, y = 12$

$(4, 4) (4, 8) \Rightarrow \text{pts.}$

If $x = 5, y = \frac{45}{4} = 11.25$

$(5, 5), (5, 10) \Rightarrow 2 \text{ pts.}$

If $x = 6, y = \frac{21}{2} = 10.5$

$(6, 6) \Rightarrow 1 \text{ pt.}$

If $x = 7, y = \frac{39}{4} = 9.75$

$(7, 7) \Rightarrow 1 \text{ pt.}$

If $x = 8, y = 9$

$(8, 8) \Rightarrow 1 \text{ pt.}$

If $x = 9, y = \frac{33}{4} = 8.25 \Rightarrow \text{no pt.}$

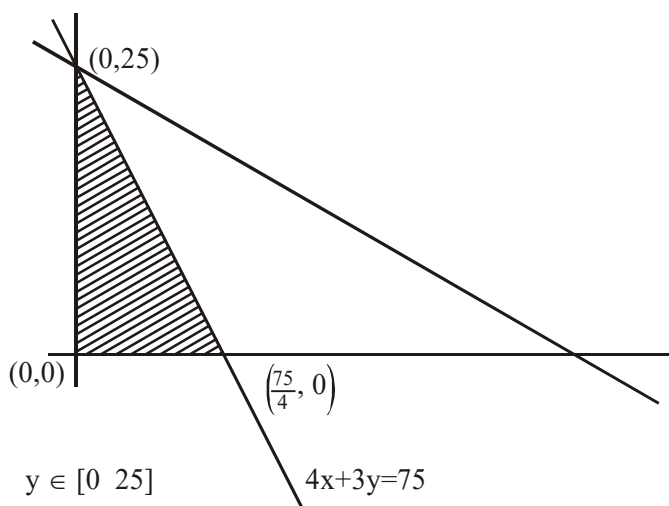
Total = 31 pts.

77. The maximum value of z in the following equation $z = 6xy + y^2$, where $3x + 4y \leq 100$ and $4x + 3y \leq 75$ for $x \geq 0$ and $y \geq 0$ is :

$z = 6xy + y^2$ का अधिकतम मान, जब कि $3x + 4y \leq 100$ तथा $4x + 3y \leq 75$, $x \geq 0, y \geq 0$ हैं, है :

Ans. 904

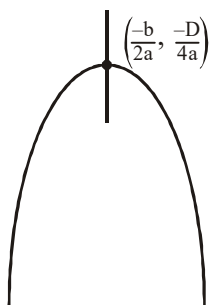
Sol.



$$z = y(6x + y) \text{ and } 4x + 3y = 75$$

$$z = \left(\frac{75 - 3y}{4} \right) 6y + y^2$$

$$z = \frac{225}{2}y - \frac{7}{2}y^2 \quad y \in [0 \ 25]$$



$$Z_{\max} = \frac{-D}{4a} = \frac{225 \times 225}{4 \times 2 \times 7} = 904.017857$$